

Sharp size thresholds for failures of log-concavity in independence systems

A seven-element obstruction, rankwise optimality, and a full-skeleton construction

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Abstract

A weak additive form of matroid augmentation does not force log-concavity. The existence of counterexamples to Xie and Xu’s rank-dependent conjecture has already been announced by Alex Chengyu Li. We therefore address a sharper extremal question: for each rank, how many ground-set elements are necessary for failure? We prove that the unrestricted minima are attained already by systems with exact augmentation parameter three. For ranks 4, 5, 6, the minimum sizes for ordinary log-concavity failure are 7, 7, 8; for every rank $r \geq 7$ the minimum is $r + 1$. Ordered log-concavity has minima 6, 7, $r + 1$ for $r = 4$, $r = 5$, and $r \geq 6$, respectively. Ground-set-normalized ultra log-concavity has minimum $r + 1$ for every $r \geq 4$.

The witnesses are a complete low-dimensional skeleton together with one simplex. We classify all three concavity properties of this family by a single coefficient inequality, establish global minimality using finite Kruskal–Katona bounds, and obtain rational generating functions, recurrences, and an exact vanishing log-concavity ratio. A standard-library Python verifier checks the displayed examples, all possible face vectors through seven active vertices, and more than two million uniform-family shadow cases. The proofs are mathematical proofs in English, not a Lean formalization. Historical priority for the refinements is not asserted.

Status and attribution. This is an unrefereed research report. It does *not* claim the first disproof of Xie–Xu Conjecture 4.8. The original conjecture, the recently located prior disproof, and the sharp extremal question investigated here are separated explicitly in Section 1. The archive contains the source, compiled article, exact verification code, and generated certificates.

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1 The problem, its current status, and the main result

1.1 Independence systems and the augmentation convention

An *independence system* is a pair $\mathcal{M} = (E, \mathcal{I})$, where E is a finite set and $\mathcal{I} \subseteq 2^E$ contains $\emptyset$ and is downward closed: if $T \in \mathcal{I}$ and $S \subseteq T$, then $S \in \mathcal{I}$. Write $n = |E|$ and

$$r = \text{rank } \mathcal{M} = \max\{|S| : S \in \mathcal{I}\}, \quad a_k = |\{S \in \mathcal{I} : |S| = k\}|.$$

Our generating polynomial is $P_{\mathcal{M}}(x) = \sum_{k=0}^n a_k x^k$. It has degree r , but we retain the zero coefficients through index n . A vertex is *active* when its singleton belongs to $\mathcal{I}$.

For a positive integer l , consider the following additive augmentation condition:

$$S, T \in \mathcal{I}, \quad |T| \geq |S| + l \implies \text{some } x \in T \setminus S \text{ satisfies } S \cup \{x\} \in \mathcal{I}. \quad (A_l)$$

Define $\lambda(\mathcal{M})$ to be the least positive l for which (A_l) holds. Such an l exists: $l = r + 1$ makes the premise impossible. The ordinary matroid augmentation axiom is A_1 . The phrase “ l -matroid” below refers only to this additive convention, not to other uses of similar terminology.

Xie and Xu introduce their terminology on printed page 14 of [1]. Their displayed axiom $(*)$ writes $x \in T$ without explicitly removing S . We require a genuinely new element, as in the usual augmentation axiom. Every counterexample below satisfies this *stronger* interpretation; thus the omission cannot invalidate the counterexamples. Exact values of λ in this report always refer to (A_l) .

1.2 The three shapes, and a normalization that matters

For a nonnegative sequence $(a_0, \dots, a_n)$ we use the following inequalities, at every $1 \leq k \leq n - 1$:

$$\text{LC :} \quad a_k^2 \geq a_{k-1}a_{k+1}, \quad (1.1)$$

$$\text{OLC :} \quad ka_k^2 \geq (k+1)a_{k-1}a_{k+1}, \quad (1.2)$$

$$\text{ULC}_n : \quad k(n-k)a_k^2 \geq (k+1)(n-k+1)a_{k-1}a_{k+1}. \quad (1.3)$$

Ordered log-concavity means log-concavity of $(k!a_k)$. Ultra log-concavity in (1.3) means log-concavity of $(a_k/\binom{n}{k})$. These equivalences follow by substituting the factorials and cancelling. Consequently ULC_n implies OLC, which implies LC. The coefficient sequence of a nonempty independence system has no internal zeros: a face of size k supplies faces of every smaller size.

The subscript n is essential. Normalizing by $\binom{r}{k}$, where r is only the degree, gives a different and stronger condition when $r < n$. The sequence in Xie–Xu Conjecture 4.8 is explicitly indexed from 0 to n , so (1.3) is the relevant normalization.

1.3 A literature-status correction and the sharper target

Xie–Xu Conjecture 4.8 asserts all three properties when

$$\lambda(\mathcal{M}) \leq \left\lfloor \frac{r}{2} \right\rfloor + 1. \quad (1.4)$$

The statement is on printed page 16 of the arXiv version [1]. It was the initial target of this investigation.

A later status check located Li's September 2026 working-paper announcement [2]. Its public overview already gives a rank-four example on thirteen elements, with sequence 1, 13, 25, 7, 2, and describes fixed-parameter families with ratios tending to zero. Thus the existence question must *not* be presented here as still open or as first resolved by this report. The full SSRN/Zenodo manuscript was not retrievable in this session; its public overview and bibliographic record were available. Detailed overlap with its unretrieved contents, and historical priority for the refinements below, remain unverified.

The resulting research target is the following exact extremal problem. It is a natural refinement, rather than a claim that the original conjecture remained unsolved at the date of this report.

Problem 1.1. For each $r \geq 4$, determine the least ground-set size at which a rank- r independence system satisfying (1.4) can fail LC, OLC, or ULC_n . Can the minima already be attained with $\lambda = 3$?

For a property $Q \in \{\text{LC}, \text{OLC}, \text{ULC}\}$, write $N_Q^{\text{all}}(r)$ for the least size of *any* rank- r independence system failing Q . Write $N_Q^{\text{XX}}(r)$ when (1.4) is imposed and $N_Q^{(3)}(r)$ when $\lambda = 3$ is imposed. In the ultra log-concave case, the normalization is always the current ground-set size.

Theorem 1.2 (Sharp rankwise thresholds). *For each $r \geq 4$ and each of the three properties,*

$$N_Q^{\text{all}}(r) = N_Q^{\text{XX}}(r) = N_Q^{(3)}(r).$$

Their common values are as follows.

Rank r	Ordinary LC	Ordered OLC	Ultra ULC_n
4	7	6	5
5	7	7	6
6	8	7	7
$r \geq 7$	$r + 1$	$r + 1$	$r + 1$

Every entry is attained by a system with all vertices active and with no vertex belonging to every maximal independent set.

The upper bounds come from one explicit family in Section 3. The lower bounds in Section 4 apply without an augmentation assumption. This is why the unrestricted and restricted minima coincide. Globally, without fixing the rank, the smallest possible ground-set sizes for failures of ordinary, ordered, and ultra log-concavity are therefore 7, 6, 5, respectively.

2 The small obstruction, with a complete direct proof

Let $E = \{1, \dots, 7\}$ and $A = \{1, 2, 3, 4\}$. Define

$$\mathcal{I} = 2^A \cup \{S \subseteq E : |S| \leq 2\}. \quad (2.1)$$

Thus every pair is independent, but the only independent sets of size three or four are the subsets of A .

Proposition 2.1. *The system in (2.1) has rank 4, exact augmentation parameter 3, and generating polynomial*

$$P(x) = 1 + 7x + 21x^2 + 4x^3 + x^4.$$

It has 34 independent sets and fails all three forms of log-concavity.

Proof. Both families in the union are downward closed. The set A is independent and no independent set has more than four elements, giving rank four. Counting at each size yields the stated coefficients and 34 total faces.

To verify A_3 , take independent S, T with $|T| \geq |S| + 3$. Since $|T| \leq 4$, we have $|S| \leq 1$. Any new element of $T \setminus S$ then produces a set of size at most two, which is independent. This proves A_3 for every possible pair S, T .

To see that A_2 fails, use $S = \{1, 5\}$ and $T = A$. Their sizes differ by two. For every $x \in T \setminus S$, the three-set $S \cup \{x\}$ contains 5, so it is not independent. Hence $\lambda = 3$. This parameter satisfies $3 = \lfloor 4/2 \rfloor + 1$.

Finally, at index 3 the ordinary inequality would require $4^2 \geq 21 \cdot 1$, whereas $16 < 21$. The stronger inequalities fail as well. $\square$

The same construction on five or six vertices separates the three properties. It is useful to see their actual, integer-valued defects. For a sequence on n elements define

$$\begin{aligned} D_{\text{LC}}(k) &= a_k^2 - a_{k-1}a_{k+1}, \\ D_{\text{OLC}}(k) &= ka_k^2 - (k+1)a_{k-1}a_{k+1}, \\ D_{\text{ULC}}(k) &= k(n-k)a_k^2 - (k+1)(n-k+1)a_{k-1}a_{k+1}. \end{aligned}$$

A negative defect is a certificate of failure.

n	Nonzero coefficients	$D_{\text{LC}}(3)$	$D_{\text{OLC}}(3)$	$D_{\text{ULC}}(3)$
5	1, 5, 10, 4, 1	6	8	-24
6	1, 6, 15, 4, 1	1	-12	-96
7	1, 7, 21, 4, 1	-5	-36	-228

Table 1: Rank-four witnesses, all with exact augmentation parameter three. All other tested indices satisfy the relevant inequalities.

For example, the five-element ultra inequality is $3 \cdot 2 \cdot 16 \geq 4 \cdot 3 \cdot 10$, or $96 \geq 120$, which is false. The six-element ordered inequality is $3 \cdot 16 \geq 4 \cdot 15$, or $48 \geq 60$, also false.

Remark 2.2 (Sparse variants). The full pair skeleton is convenient, not required. From the five-element example delete the pair $\{1, 5\}$; from the six-element example delete $\{1, 5\}, \{5, 6\}$; from the seven-element example delete $\{1, 5\}, \{5, 6\}, \{5, 7\}, \{6, 7\}$. The resulting sequences are

$$(1, 5, 9, 4, 1), \quad (1, 6, 13, 4, 1), \quad (1, 7, 17, 4, 1).$$

They have 20, 25, 30 faces and still have $\lambda = 3$. Indeed, the only nontrivial A_3 test beyond the empty set has a singleton S and $T = A$; every outside vertex still has a permitted pair with A . The same A_2 obstruction can be chosen from a retained mixed pair. The last sequence has the unit ordinary defect $16 - 17 = -1$. We do not need a claim of minimum total face count for the rankwise ground-size theorem.

3 The full-skeleton-plus-simplex family

Fix integers $n > r \geq 2$ and $1 \leq s \leq r - 1$, with $A \subset E$, $|A| = r$, $|E| = n$. Put

$$\mathcal{I}_{n,r,s} = 2^A \cup \{S \subseteq E : |S| \leq s\}. \quad (3.1)$$

The word “skeleton” will refer to the cutoff by *cardinality* s ; in the usual topological dimension convention it is the $(s - 1)$ -skeleton.

Theorem 3.1 (Counts and exact augmentation). *The system (3.1) has rank r , has no inactive vertices, and satisfies*

$$a_k = \begin{cases} \binom{n}{k}, & 0 \leq k \leq s, \\ \binom{r}{k}, & s + 1 \leq k \leq r, \\ 0, & r < k \leq n. \end{cases} \quad \lambda(\mathcal{I}_{n,r,s}) = r - s + 1. \quad (3.2)$$

Its maximal independent sets are A and the s -sets not contained in A . No vertex belongs to all maximal independent sets.

Proof. The downward-closure and rank statements follow directly from the construction; counting gives (3.2). Let $l = r - s + 1$. If $|T| \geq |S| + l$, then $|S| \leq r - l = s - 1$. Thus adjoining any element of $T \setminus S$ produces a set of size at most s . This verifies A_l .

Choose an s -set S meeting $E \setminus A$ and put $T = A$. The size difference is $r - s$. Every possible new element gives a mixed $(s + 1)$ -set, absent from (3.1). Hence A_{r-s} fails, proving the exact parameter.

An independent set contained in A extends to A . A mixed independent set extends to a mixed s -set, and such an s -set cannot be extended. This proves the description of maximal sets. Any outside vertex is omitted by A . Given $v \in A$, choose a mixed s -set omitting v : there are at least r other available elements and $s \leq r - 1$. This maximal set omits v , completing the proof. $\square$

A vertex belongs to every maximal independent set if and only if it can be adjoined to every independent set. To check the nontrivial direction, extend an arbitrary face to a maximal face containing the vertex, and use downward closure. Such a vertex is often called a cone vertex or a coloop in this setting. Thus the theorem excludes a construction obtained merely by adjoining universal independent elements.

When $s \geq 2$, every pair is a face, so the one-skeleton is complete and the simplicial complex is connected. For $s = r - 2$ and $r \geq 4$ it is not the independence complex of a simple graph: all pairs are independent, but some larger sets are not. Graph-realizable examples will be treated separately.

3.1 Exactly one inequality can fail

Theorem 3.2 (Complete classification within the family). *For $1 \leq s \leq r - 2$, put*

$$L = \binom{n}{s}, \quad M = \binom{r}{s+1}, \quad H = \binom{r}{s+2}.$$

The entire sequence (3.2) is

$$\text{LC} \iff M^2 \geq LH, \quad (3.3)$$

$$\text{OLC} \iff (s+1)M^2 \geq (s+2)LH, \quad (3.4)$$

$$\text{ULC}_n \iff (s+1)(n-s-1)M^2 \geq (s+2)(n-s)LH. \quad (3.5)$$

In each case the only possible failing index is $s + 1$. For $s = r - 1$, all three properties hold for every $n > r$.

Proof. For a positive sequence, log-concavity is equivalent to its successive ratios being nonincreasing. In a binomial block, the three relevant ratio sequences are obtained from

$$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}, \quad \frac{(k+1)!\binom{n}{k+1}}{k!\binom{n}{k}} = n-k, \quad \frac{\binom{n}{k}}{\binom{n}{k}} = 1.$$

They show all three inequalities strictly below the transition. For the upper block, ordinary and ordered ratios are respectively $(r-k)/(k+1)$ and $r-k$, again decreasing. The ultra-normalized upper block has ratios

$$\frac{\binom{r}{k+1}/\binom{n}{k+1}}{\binom{r}{k}/\binom{n}{k}} = \frac{r-k}{n-k}. \quad (3.6)$$

These decrease with k , since the difference between consecutive values has numerator $n-r > 0$.

At index s , the middle and previous terms are those of the lower binomial block, while the next term has only decreased from $\binom{n}{s+1}$ to $\binom{r}{s+1}$. The desired inequalities are therefore preserved. Every index at least $s+2$ lies entirely in the upper block, or meets the terminal zeros, and has already been covered. Substitution at index $s+1$ gives exactly (3.3)–(3.5).

When $s = r - 1$, the sole transition is a final downward change in a_r ; there is no following positive coefficient capable of making an inequality at $s+1$ fail. The same block argument proves all three properties. $\square$

3.2 Parameter three and the explicit thresholds

The decisive specialization is $s = r - 2$, for $r \geq 3$. It gives

$$P_{n,r}(x) = \sum_{k=0}^{r-2} \binom{n}{k} x^k + r x^{r-1} + x^r, \quad \lambda = 3. \quad (3.7)$$

All three global tests reduce to the last nontrivial index $r - 1$:

$$\text{LC} \iff \binom{n}{r-2} \leq r^2, \quad (3.8)$$

$$\text{OLC} \iff \binom{n}{r-2} \leq r(r-1), \quad (3.9)$$

$$\text{ULC}_n \iff \binom{n}{r-2} \leq r(r-1) \frac{n-r+1}{n-r+2}. \quad (3.10)$$

For every $r \geq 4$, parameter three is allowed by (1.4). These are necessary and sufficient tests for the full sequence, not just sufficient tests for failure.

The codimension-one case, $n = r + 1$, is particularly economical. Let $Q_r = P_{r+1,r}$. Then

$$Q_r(x) = (1+x)^{r+1} - \binom{r}{2} x^{r-1} - r x^r - x^{r+1}. \quad (3.11)$$

Since $\binom{r+1}{r-2} = \binom{r+1}{3}$, cancellation in (3.8)–(3.10) gives the exact classification

Property of Q_r ($r \geq 3$)	Ranks for which it holds
LC	$3 \leq r \leq 6$
OLC	$3 \leq r \leq 5$
ULC_{r+1}	$r = 3$.

(3.12)

For ordinary log-concavity, for example, the test becomes $r^2 - 6r - 1 \leq 0$, whose integral solutions with $r \geq 3$ are precisely 3, 4, 5, 6. For ordered log-concavity it becomes $r + 1 \leq 6$. For ultra log-concavity it becomes $r + 1 \leq 4$.

4 Global minimality, not just minimality within the construction

A counterexample does not by itself show that it is small in an absolute sense. We now give finite, exact lower-bound certificates for all independence systems needed to prove [theorem 1.2](#).

4.1 The shadow inequality

For a family $\mathcal{F}$ of q -element sets, its *lower shadow* $\partial\mathcal{F}$ consists of all $(q-1)$ -sets contained in at least one member. If $m > 0$, write its unique canonical q -binomial expansion

$$m = \binom{b_q}{q} + \binom{b_{q-1}}{q-1} + \cdots + \binom{b_j}{j}, \quad b_q > b_{q-1} > \cdots > b_j \geq j \geq 1. \quad (4.1)$$

Define

$$\sigma_q(m) = \binom{b_q}{q-1} + \binom{b_{q-1}}{q-2} + \cdots + \binom{b_j}{j-1}, \quad \sigma_q(0) = 0. \quad (4.2)$$

The Kruskal–Katona shadow theorem gives $|\partial\mathcal{F}| \geq \sigma_q(|\mathcal{F}|)$. A primary-source account of the shadow theorem and its colex form appears in the introduction to Bukh’s paper [\[3\]](#); the original references are [\[4, 5\]](#).

For an independence system, every shadow face is independent. Hence

$$\sigma_{k+1}(a_{k+1}) \leq a_k. \quad (4.3)$$

The expansion and its shadow can be evaluated by successive greedy choices of the largest possible top index. For example,

$$26 = \binom{6}{3} + \binom{4}{2}, \quad \sigma_3(26) = \binom{6}{2} + \binom{4}{1} = 19.$$

This example is also one of the boundary values in the seven-vertex enumeration in the archive.

4.2 A small table that proves the lower bounds

For a fixed n and index k , let

$$U_{n,k}(b) = \max\{c : 0 \leq c \leq \binom{n}{k+1}, \quad \sigma_{k+1}(c) \leq b\}. \quad (4.4)$$

If $a_k = b > 0$, then

$$\frac{a_{k-1}a_{k+1}}{a_k^2} \leq \frac{\binom{n}{k-1}U_{n,k}(b)}{b^2}. \quad (4.5)$$

Multiply the right side by 1, $(k+1)/k$, or $(k+1)(n-k+1)/(k(n-k))$ for ordinary, ordered, or ultra log-concavity. A maximum of at most one proves the corresponding inequality for *every* independence system under consideration.

For active ground sets, at $k = 1$ the value of b is fixed to n . At $k \geq 2$, maximize over $1 \leq b \leq \binom{n}{k}$. Under the extra assumption $\text{rank } \mathcal{M} \geq t$, every k -subset of some t -face is independent; therefore one may restrict to

$$\max\{1, \binom{t}{k}\} \leq b \leq \binom{n}{k}. \quad (4.6)$$

Table 2 records the resulting exact maxima. A dash denotes an index outside the sequence.

Case	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
$n = 6$, LC	5/12	2/3	15/16	4/5	5/12	—
$n = 5$, OLC	4/5	5/6	5/6	1/2	—	—
$n = 4$, ULC ₄	1	1	1	—	—	—
$n = 7$, $r \geq 6$, LC	3/7	28/45	63/80	14/15	35/36	3/7
$n = 6$, $r \geq 5$, OLC	5/6	9/10	1	1	1/2	—

Table 2: Finite shadow bounds proving universal concavity in the indicated cases. Every entry is at most one. For ULC₂ and ULC₃, the analogous entries are all one. All maximizing pairs (b, c) are in `data/minimality_bounds.csv`.

These are short finite arithmetic calculations, not probabilistic tests. For example, the 15/16 entry is the maximum of $15U_{6,3}(b)/b^2$ over $1 \leq b \leq 20$, and is attained at $b = 4$, $c = 1$. For $n = 7$, $r \geq 6$, and $k = 5$, one has $b \geq \binom{6}{5} = 6$; the maximum is $\binom{7}{4}/6^2 = 35/36$ at $c = 1$. The complete algorithm defining every entry is (4.1)–(4.6), with no omitted search range. Appendix A and the verifier provide two ways to reproduce the finite calculations.

Zero middle coefficients cause no problem. Downward closure then implies $a_{k+1} = 0$, so all the relevant inequalities at k hold. Inactive ground elements can also be handled exactly. Removing them leaves ordinary and ordered inequalities unchanged. For ultra log-concavity, if the active size is $m \leq n$, the factor

$$\frac{k+1}{k} \frac{n-k+1}{n-k}$$

is no larger than the same factor with m in place of n whenever $k < m$. Thus ULC _{m} on the active set implies ULC _{n} after adding inactive elements. For active sizes 1, 2, 3, 4 the ultra bounds are vacuous or the all-one rows described in table 2.

For ordinary and ordered lower bounds on ground sets smaller than the first two rows, one can simply pad with inactive elements up to six or five. The $k \geq 2$ bounds remain valid because they use only the upper ambient size; the index 1 inequality follows directly from $a_2 \leq \binom{a_1}{2}$. In the two rank-constrained rows, deleting an inactive element leaves a full simplex when the rank equals the remaining active size, which satisfies all the required inequalities.

Proposition 4.1 (Universal small-size obstructions). *Every independence system on at most six elements is ordinarily log-concave. Every one on at most five elements is ordered log-concave. Every one on at most four elements is ultra log-concave with its actual ground-set normalization. Moreover,*

every rank-six system on seven elements is ordinarily log-concave, and every rank-five system on six elements is ordered log-concave.

Proof. Apply the at-most-one bounds in table 2, with the preceding zero and inactive-element observations. $\square$

4.3 Completion of the sharp rankwise theorem

Proof of theorem 1.2. Any rank- r system has $n \geq r$. If $n = r$, an independent r -set must be the whole ground set, and downward closure forces the full simplex. Its binomial sequence has all three properties. Therefore all desired minima are at least $r + 1$.

For ordinary log-concavity, theorem 4.1 increases this lower bound to seven when $r = 4$ or 5, and to eight when $r = 6$. For ordered log-concavity, it increases the bound to six for $r = 4$ and to seven for $r = 5$. No additional lower bound is needed for the other entries.

For the matching upper bounds, use $\mathcal{I}_{n,r,r-2}$. When $r = 4$, choose $n = 7, 6, 5$ for the three respective failures, as in table 1. For $r = 5$, the ordinary and ordered failures at $n = 7$ follow from $\binom{7}{3} = 35 > 25 > 20$; ultra failure occurs at $n = 6$ by (3.12). For $r = 6$, ordinary failure at $n = 8$ follows from $\binom{8}{4} = 70 > 36$; the other two failures at $n = 7$ again follow from (3.12). Every remaining entry is the codimension-one case of that classification.

All these systems have $\lambda = 3$, all vertices active, and no coloops by theorem 3.1. For $r \geq 4$, exact parameter three satisfies (1.4). The classes are nested, so their minima satisfy

$$N_Q^{\text{all}}(r) \leq N_Q^{\text{XX}}(r) \leq N_Q^{(3)}(r).$$

The unrestricted lower bounds and parameter-three witnesses force equality throughout. $\square$

5 Sequences, generating functions, and asymptotic strength

5.1 The extremal size sequences

The main theorem produces three elementary integer sequences, indexed by $r \geq 4$:

r	4	5	6	7	8	9	10	11
$N_{\text{LC}}(r)$	7	7	8	8	9	10	11	12
$N_{\text{OLC}}(r)$	6	7	7	8	9	10	11	12
$N_{\text{ULC}}(r)$	5	6	7	8	9	10	11	12

Their usefulness is their extremal interpretation; no claim of a new numerical sequence or an existing OEIS accession is made. Put $G_Q(z) = \sum_{j \geq 0} N_Q(j+4)z^j$. Summing the eventual arithmetic progression gives

$$G_{\text{ULC}}(z) = \frac{5 - 4z}{(1 - z)^2}, \quad (5.1)$$

$$G_{\text{OLC}}(z) = \frac{5 - 4z}{(1 - z)^2} + 1 + z, \quad (5.2)$$

$$G_{\text{LC}}(z) = \frac{5 - 4z}{(1 - z)^2} + 2 + z + z^2. \quad (5.3)$$

Thus the second differences vanish beyond the finitely many exceptional ranks. These identities are formal power-series identities; analytically they hold for $|z| < 1$.

5.2 Counting all faces in the codimension-one family

Let $N_r = Q_r(1)$, where Q_r is given by (3.11). For $r \geq 3$,

$$N_r = 2^{r+1} - 1 - \binom{r+1}{2}. \quad (5.4)$$

Indeed, the excluded faces are $\binom{r}{2}$ sets at size $r-1$, r sets at size r , and one set at size $r+1$. Their total is $1 + \binom{r+1}{2}$.

Extend the formula algebraically to $r = 0, 1, 2$; no member of (3.1) with cutoff $r-2$ is being claimed at these indices. The resulting sequence begins

$$1, 2, 4, 9, 21, 48, 106, 227, 475, 978, 1992, \dots$$

Since $\sum_{r \geq 0} \binom{r+1}{2} z^r = z/(1-z)^3$, we obtain

$$\sum_{r \geq 0} N_r z^r = \frac{2}{1-2z} - \frac{1}{1-z} - \frac{z}{(1-z)^3} = \frac{1-3z+3z^2}{(1-2z)(1-z)^3}. \quad (5.5)$$

Consequently

$$N_{r+4} = 5N_{r+3} - 9N_{r+2} + 7N_{r+1} - 2N_r \quad (r \geq 0). \quad (5.6)$$

This follows either by multiplying (5.5) by its denominator or by noting that 2^r and every polynomial of degree at most two are annihilated by the characteristic polynomial $(t-2)(t-1)^3$.

5.3 A failure ratio tending to zero

For the codimension-one sequence, the ratio at the failing position is exactly

$$R_r = \frac{a_{r-1}^2}{a_{r-2}a_r} = \frac{r^2}{\binom{r+1}{3}} = \frac{6r}{r^2-1}. \quad (5.7)$$

Thus $R_r < 1$ exactly when $r \geq 7$, and $R_r \rightarrow 0$. More explicitly, for every $J \geq 0$ and $r > 1$,

$$R_r = 6 \sum_{j=0}^J r^{-(2j+1)} + \frac{6r^{-(2J+3)}}{1-r^{-2}}. \quad (5.8)$$

The remainder is positive and exact, so the asymptotic expansion $6/r + 6/r^3 + 6/r^5 + \dots$ needs no unquantified error estimate. For $r \geq 2$, the displayed remainder is at most $8r^{-(2J+3)}$.

At the same time, $n = r+1$, $\lambda = 3$, every singleton is independent, and there are no coloops. In particular,

$$\frac{r}{n} \rightarrow 1, \quad \frac{\lambda}{r} \rightarrow 0.$$

The coefficient defect is the cubic integer expression

$$a_{r-2}a_r - a_{r-1}^2 = \frac{r^3 - 6r^2 - r}{6}, \quad (5.9)$$

which starts 7, 20, 39, 65, ... at $r = 7, 8, 9, 10, \dots$. Its fourth finite difference vanishes.

Corollary 5.1. *No positive constant c can make $a_k^2 \geq c a_{k-1} a_{k+1}$ hold uniformly for every independent-set sequence with $\lambda = 3$, even after requiring all vertices active, no coloops, and $n = r + 1$. Nor can an upper bound on λ depending only on rank restore ordinary log-concavity if that bound permits $\lambda = 3$ at arbitrarily large ranks.*

Proof. Use (5.7) at the corresponding arbitrarily large ranks. $\square$

These qualitative phenomena overlap with those already advertised in Li's overview [2]. The contribution established here is the explicit codimension-one construction, the sharp shape tests, and their use in the exact ground-size theorem; this paragraph is not a priority claim for the qualitative nonexistence of a uniform bound.

6 What the augmentation parameter actually measures

The following structural description is useful for understanding why a rank-dependent hypothesis is too weak. It also explains the direct-sum method mentioned in the prior-work overview.

For $U \subseteq E$, restrict the system to U and let $\mathcal{B}(U)$ be its inclusion-maximal independent sets. Define

$$w(U) = \max_{B \in \mathcal{B}(U)} |B| - \min_{B \in \mathcal{B}(U)} |B|.$$

In particular $w(\emptyset) = 0$.

Proposition 6.1 (Hereditary maximal-set width). *For every finite independence system,*

$$\lambda(\mathcal{M}) = 1 + \max_{U \subseteq E} w(U). \quad (6.1)$$

Proof. Let W be the maximum on the right. Choose U and maximal independent sets B, C of minimum and maximum size in U , so that $|C| - |B| = W$. When $W \geq 1$, no element of $C \setminus B$ can extend B , because B is already maximal in U . Thus A_W fails and $\lambda \geq W + 1$. For $W = 0$, the same lower bound is simply $\lambda \geq 1$.

Conversely, suppose independent S, T admit no extension of S by any new element of T . In the restriction to $U = S \cup T$, the set S is maximal: otherwise a larger face containing it would supply a one-element extension by downward closure. Extend T to a maximal face C in U . Then

$$|T| - |S| \leq |C| - |S| \leq w(U) \leq W.$$

No unaugmentable pair can therefore have a size gap at least $W + 1$. This is A_{W+1} and gives the reverse inequality. $\square$

Thus $\lambda = 3$ controls the spread of maximal-set sizes in every restriction by two. It does not control the number of faces at adjacent sizes. The family (3.1) has an enormous collection of lower faces and only the faces of one simplex above the cutoff; the width condition alone cannot prevent that imbalance.

If systems on disjoint ground sets have independent families $\mathcal{I}_1, \mathcal{I}_2$, their direct sum has independent sets $S_1 \cup S_2$ with $S_i \in \mathcal{I}_i$. Maximal faces in each restriction are formed independently in the two components, so their widths add. Taking maxima in (6.1) yields the exact identity

$$\lambda(\mathcal{M}_1 \oplus \mathcal{M}_2) = \lambda(\mathcal{M}_1) + \lambda(\mathcal{M}_2) - 1. \quad (6.2)$$

Ranks add and generating polynomials multiply. In particular, a matroid factor preserves the augmentation parameter of the other factor. This identity explains how augmentation defects can remain fixed while rank increases; it is not being advertised as a newly discovered method relative to Li's prior overview.

7 Graph independence polynomials and removal of isolated vertices

The principal minimal examples are not graph independence complexes. It is nonetheless useful to verify that failure can occur in the graph subclass too, without confusing independence and dependence polynomials. For a graph G , let $I(G, x)$ count vertex sets containing no edge.

Let $H_m = K_m \vee \overline{K}_3$ be the join of a clique of size m with three mutually nonadjacent vertices: every vertex of the clique is joined to each of those three vertices. A nonempty independent set is either a subset of the three vertices or one clique vertex. Hence

$$I(H_m, x) = (1 + x)^3 + mx. \quad (7.1)$$

For $m \geq 1$, its independence system has exact parameter three. Its rank is three, a clique singleton cannot be extended toward the independent triple, and A_3 has only the empty-set case to check.

7.1 Arbitrarily large rank with a simple tail certificate

Add $c = r - 3$ isolated vertices. Their independence system is a free matroid, so (6.2) preserves parameter three. The new polynomial is

$$F_{r,m}(x) = (1 + x)^r + mx(1 + x)^{r-3}. \quad (7.2)$$

Its top three coefficients are

$$a_{r-2} = \binom{r}{2} + m, \quad a_{r-1} = r, \quad a_r = 1.$$

The last ordinary log-concavity inequality fails precisely when $m > \binom{r+1}{2}$. In particular, choosing $m = \binom{r+1}{2} + 1$ gives the tail $(r^2 + 1, r, 1)$ and unit defect -1 . The graph has $r + m = \binom{r+2}{2}$ vertices. No minimum-graph-size claim is made here.

7.2 No isolated vertices and no cone vertices

Instead of isolated vertices, take $c \geq 1$ disjoint cliques of size $t \geq 2$. Each clique contributes a rank-one matroid with polynomial $1 + tx$. Thus

$$G_{c,m,t} = H_m \sqcup cK_t, \quad I(G_{c,m,t}, x) = ((1 + x)^3 + mx)(1 + tx)^c \quad (7.3)$$

has rank $r = c + 3$, exact parameter three, and no isolated graph vertices. A graph vertex can be adjoined to every independent set exactly when it is isolated, so these complexes also have no cone vertices. The graph itself is generally disconnected; that is a different notion from connectedness of the simplicial complex discussed earlier.

Reading the last three coefficients of (7.3) gives

$$\begin{aligned} a_r &= t^c, \\ a_{r-1} &= 3t^c + ct^{c-1}, \\ a_{r-2} &= (3+m)t^c + 3ct^{c-1} + \binom{c}{2}t^{c-2}. \end{aligned}$$

Since $t > 0$, these expressions are valid also when $c = 1$ (the last term has zero coefficient). Direct expansion yields

$$a_{r-1}^2 - a_{r-2}a_r = t^{2c} \left(6 - m + \frac{3c}{t} + \frac{c(c+1)}{2t^2} \right). \quad (7.4)$$

For instance $c = 1, m = 8, t = 2$ produces 1, 13, 25, 7, 2, the same sequence advertised in Li's prior overview, with $7^2 - 25 \cdot 2 = -1$.

For a uniform arbitrary-rank choice, set $m = 7$ and $t = 4c$. The factor in parentheses is

$$-1 + \frac{3}{4} + \frac{c+1}{32c} \leq -\frac{3}{16} < 0.$$

This gives rank- r examples with no isolated vertices on $10 + 4(r-3)^2$ vertices for every $r \geq 4$. These formulas provide independent, elementary verification of the graph variants; they do not assert priority over the prior direct-sum method.

8 A real-rootedness obstruction and the two normalizations

Even a member of (3.7) that satisfies ULC_n need not be real-rooted. There is a short proof requiring only differentiation and Rolle's theorem.

Proposition 8.1. *For every $n > r \geq 3$, the polynomial $P_{n,r}$ has at least one nonreal root.*

Proof. If all roots were real, repeated differentiation would preserve that property. The derivative of order $r-2$ is the quadratic

$$P_{n,r}^{(r-2)}(x) = (r-2)! \binom{n}{r-2} + r(r-1)!x + \frac{r!}{2}x^2.$$

Its discriminant is

$$r(r-1)(r-2)!^2 \left(r(r-1) - 2 \binom{n}{r-2} \right).$$

A nonnegative discriminant would require $\binom{n}{r-2} \leq r(r-1)/2 = \binom{r}{r-2}$, which is impossible because $n > r$ and $r-2 \geq 1$. $\square$

For example, the $r = 3, n = 4$ polynomial $1 + 4x + 3x^2 + x^3$ satisfies ULC_4 by (3.12), yet it has nonreal roots by the proposition. This does not conflict with any implication from real-rootedness to normalized coefficient inequalities. Normalization by the degree r is stronger than normalization by the ambient n : at the last index of (3.7), the degree-normalized ultra inequality would require the very bound $\binom{n}{r-2} \leq \binom{r}{2}$ just ruled out.

9 Exact verification, reproducibility, and remaining scope

9.1 What was actually executed

The archive's `verify.py` uses only the Python standard library. Its full run completed successfully with Python 3.13.5. All comparisons use integers or exact rational numbers; no numerical root approximation or floating-point sign decision supports a theorem.

The verifier produced the counts below; their JSON summary is included in the `data/` directory:

Check	Cases
General family shape classifications, $n \leq 24$	2,024
Explicit face-system and augmentation checks, $n \leq 8$	56
Direct-sum parameter checks	6
Possible face vectors on 1 through 7 active vertices	5,459
Finite maximal-ratio certificates	26
Graph polynomial identities	54
Unrestricted uniform-family shadow cases, $n \leq 6$	1,116,354
Core-constrained uniform-family cases	1,116,352

The last two rows total 2,232,706 uniform-family cases. These are not claimed to be that many distinct independence systems.

The core-constrained check independently examines all uniform families containing the faces of a fixed six-element core on seven vertices, and of a fixed five-element core on six vertices, at the levels needed by the two exceptional rank lower bounds. It avoids enumerating all labelled complexes on seven vertices. The largest optional uniform layer has 20 members, so this independent shadow check has at most 2^{20} states in a single layer.

The separate face-vector enumeration uses the Kruskal–Katona criterion. For each possible a_k , it recursively accepts exactly the values with $\sigma_k(a_k) \leq a_{k-1}$. These inequalities are sufficient as well as necessary: choose the initial colex segment at each level; its shadow is an initial colex segment of the preceding level, and the inequalities ensure containment. The enumeration therefore concerns possible *vectors*, not just hypothetical arrays, while distinct complexes with the same vector are counted once.

Active vertices	Possible vectors	Fail LC	Fail OLC	Fail ULC _n
1	1	0	0	0
2	2	0	0	0
3	5	0	0	0
4	16	0	0	0
5	70	0	0	5
6	457	0	3	47
7	4,908	23	120	809

These counts are useful checks but are not substituted for the stated rankwise theorem. In particular, counts of shape failures do not by themselves assert that all failing systems obey A_3 .

9.2 Reproduction

From the archive root, run

```
python verify.py --full --output regenerated-data
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the mathematical certificates. It records the local Python version, so that metadata field may differ across machines while the mathematical data agree. The LaTeX file is self-contained, including its bibliography, and needs no BibTeX run. The accompanying `README.md` describes the files, bit-mask convention, and expected output.

9.3 What is and is not settled

The exact extremal problem theorem 1.1 is resolved by theorem 1.2, with explicit witnesses and universal lower bounds. The three-parametric family’s concavity classification and all asymptotic claims have proofs independent of the computations. The finite lower bounds use the classical shadow theorem plus the completely specified finite arithmetic table; the exhaustive uniform-family checks give a second verification for the specific sizes needed.

This does not prove or disprove the parameter-two conjecture: every main counterexample has *exact* parameter three. Xie and Xu separately formulate the parameter-two assertion as Conjecture 4.4 in [1]. The present construction with $s = r - 1$ has parameter two and satisfies all three shapes, but it is only a subclass of parameter-two systems. Nor does an example about a graph’s *independent* sets refute the same paper’s Conjecture 4.1 about its *dependent* sets. These are separate counting problems.

Finally, the original existence disproof is prior work, not a new result claimed here. The search was not a complete priority audit, and the unretrieved full Li manuscript leaves possible overlap in refinements unresolved. The strongest unconditional output of this report is the proved sharp theorem and its reproducible certificates, not a claim of publication priority or independent peer review.

A Finite arithmetic certificates and independent shadow checks

A.1 The entire maximal-ratio calculation

The following pseudocode is the finite proof behind each entry of table 2. The function `shadow` is the explicit binomial expansion (4.1) followed by (4.2); it does not call an optimization oracle.

```
for k in range(1, n):
    allowed_b = [n] if k == 1 else range(max(1, binom(t, k)),
                                         binom(n, k) + 1)

    maximum = 0
    for b in allowed_b:
        c = max(c for c in range(binom(n, k + 1) + 1)
                if shadow(c, k + 1) <= b)
        value = factor(kind, n, k) * binom(n, k - 1) * c / b**2
        maximum = max(maximum, value)
    assert maximum <= 1
```

Here $t = 0$ means no rank lower bound. The actual program uses `Fraction`, and its checking function remains active under `python -O`. The displayed divisions in the pseudocode must not be implemented with floating-point arithmetic when an exact certificate is desired.

For illustration, the complete shadow minima for $q = 4$ on six vertices, indexed by the number $m = 0, 1, \dots, 15$ of four-sets, are

$$0, 4, 7, 9, 10, 10, 13, 15, 16, 16, 18, 19, 19, 20, 20, 20.$$

Thus one four-set forces four triples, five four-sets force ten triples, and six four-sets force thirteen triples. Such small discontinuities are why the exact shadow bound is substantially sharper than merely counting incidences between adjacent levels.

A.2 An independent enumeration of lower shadows

Represent every subset of $\{1, \dots, n\}$ by an integer mask, with vertex j represented by bit $j - 1$. First enumerate all q -subsets and all $(q - 1)$ -subsets. A q -set then has its own lower-shadow bit mask in the $(q - 1)$ -set universe. For a family mask $F \neq 0$, remove its lowest set bit b and use

$$\text{shadowMask}(F) = \text{shadowMask}(F \setminus \{b\}) \text{ OR } \text{shadowMask}(\{b\}).$$

Group the resulting popcounts by $|F|$ and take minima. This is independent of the canonical binomial expansion. For all unrestricted layers on at most six vertices, the computed minima agree exactly with $\sigma_q(m)$.

For a fixed core, initialize the shadow mask with all its required lower faces and enumerate only the optional upper faces. Taking the minimum lower count for each optional upper count yields a direct finite check of the two exceptional rank rows. Every rank- r system on $r + 1$ vertices contains an r -face; relabel it as the prescribed core. Therefore the core enumeration covers every system needed for those two rows, even though it does not impose all compatibility conditions between optional layers. Omitting those extra conditions only enlarges the tested class, which is safe for a universal lower bound.

B Compact classification of the smallest failing face vectors

The following tables record all failing vectors at the three first possible sizes, before any augmentation restriction is imposed. Terminal zeros through the ambient size are suppressed.

At five vertices, the ultra log-concavity failures are

$$(1, 5, 3, 1), \quad (1, 5, 6, 4), \quad (1, 5, 6, 4, 1), \quad (1, 5, 9, 4, 1), \quad (1, 5, 10, 4, 1).$$

At six vertices, the ordered failures are exactly

$$(1, 6, b, 4, 1), \quad b = 13, 14, 15.$$

The full list of twenty-three ordinary failures on seven vertices can be compressed as follows:

Vector pattern	Parameter range	Number
$(1, 7, b, 4, 1)$	$17 \leq b \leq 21$	5
$(1, 7, 21, 10, 5)$	—	1
$(1, 7, 21, 10, 5, 1)$	—	1
$(1, 7, 19, 26, 5, 1)$	—	1
$(1, 7, 20, c, 5, 1)$	$26 \leq c \leq 30$	5
$(1, 7, 21, c, 5, 1)$	$26 \leq c \leq 35$	10

The list immediately provides another check that none of the seven-vertex ordinary failures has rank six, and none of the six-vertex ordered failures has rank five. These lists are consequences of the finite face-vector criterion, not a classification of labelled systems or their augmentation parameters.

References

- [1] Yan-Ting Xie and Shou-Jun Xu, *Ultra log-concavity and real-rootedness of dependence polynomials*, arXiv:2408.09152v1, August 17, 2024. [arXiv record](#). The relevant locations are the sequence definitions on printed page 3, the additive augmentation condition on page 14, and Conjecture 4.8 on page 16. The record was checked on September 19, 2026.
- [2] Alex Chengyu Li, *Additive Augmentation Gaps Do Not Force Log-Concavity*, working paper, 2026. DOI: [10.2139/ssrn.7434141](#). Author’s public overview: [Crab Research, augmentation-gap-log-concavity](#). That page reports first public release September 6, 2026 and revision September 18, 2026. Overview accessed September 19, 2026; the full linked manuscript was not retrieved in this investigation.
- [3] Boris Bukh, *Multidimensional Kruskal–Katona theorem*, arXiv:1009.2375v2, 2011. [arXiv record](#). The introduction states the classical colex shadow theorem and the Lovász form, and gives the original references below.
- [4] Joseph B. Kruskal, *The number of simplices in a complex*, in *Mathematical Optimization Techniques*, University of California Press, Berkeley, 1963, pp. 251–278.
- [5] Gyula Katona, *A theorem of finite sets*, in *Theory of Graphs* (Proceedings of the Colloquium, Tihany, 1966), Academic Press, New York, 1968, pp. 187–207.